

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : MCQ Test (Low)
Assessment Tool Weightage: 30%

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	5
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	5
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	5
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	5
5	Functional dependency $X \rightarrow Y$ means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	5
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	5
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An attribute that depends on a non-prime attribute (d) A foreign key reference	BL2 – Understand	5

8	If R(A,B,C) has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	5
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	5
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	5

Code of Conduct:

*I G Vyshnavi (192525260)_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:vyshnavi

Question 1

Which of the following is true about 1NF?

- (a) Allows multi-valued attributes
- (b) Eliminates partial dependencies
- (c) All attributes must be atomic
- (d) Eliminates transitive dependencies

Correct Answer: (c)

Question 2

A relation is in 2NF if it is in 1NF and:

- (a) Has no transitive dependencies
- (b) Has no partial dependencies on primary key
- (c) Every attribute is a prime attribute
- (d) Has no multi-valued dependencies

Correct Answer: (b)

Question 3

Which normal form deals with multi-valued dependencies?

- (a) 2NF
- (b) 3NF
- (c) BCNF
- (d) 4NF

Correct Answer: (d)

Question 4

BCNF is a stricter version of:

- (a) 1NF
- (b) 2NF
- (c) 3NF
- (d) 4NF

Correct Answer: (c)

Question 5

Functional dependency $X \rightarrow Y$ means:

- (a) Y determines X
- (b) X determines Y
- (c) X and Y are unrelated
- (d) Both determine each other

Correct Answer: (b)

Question 6

Which of the following anomaly is NOT associated with unnormalized relations?

- (a) Insertion anomaly
- (b) Deletion anomaly
- (c) Update anomaly
- (d) Retrieval anomaly

Correct Answer: (d)

Question 7

A superkey is:

- (a) A minimal set of attributes that uniquely identifies tuples
- (b) Any set of attributes that uniquely identifies tuples
- (c) An attribute that depends on a non-prime attribute
- (d) A foreign key reference

Correct Answer: (b)

Question 8

If $R(A,B,C)$ has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is:

- (a) Partially dependent on A
- (b) Fully dependent on A
- (c) Transitively dependent on A

(d) Independently determined

Correct Answer: (c)

Question 9

5NF is also known as:

(a) Domain-Key Normal Form

(b) Boyce-Codd Normal Form

(c) Project-Join Normal Form

(d) Element Normal Form

Correct Answer: (c)

Question 10

Lossless-join decomposition ensures:

(a) No data is lost during decomposition

(b) All FDs are preserved

(c) The relation is in BCNF

(d) No null values exist

Correct Answer: (a)